Practical - 8

MySQL Schema Design, ER Model, PDO Connectivity, and Prepared Statements

Problem Definition Design the database for StudentHub with students, events and registrations. Create PHP database connection using MYSQL statements.

A. Theory

Is the schema normalized properly?

Yes. The schema is properly structured into separate tables such as students, events, and registrations, which reduces duplicate data. The registrations table handles the many-to-many relationship between students and events.

Are primary keys and foreign keys correctly defined?

Yes. Each table has its own Primary Key, such as student_id, event_id, and registration_id. The registrations table uses student_id and event_id as Foreign Keys to connect the related tables.

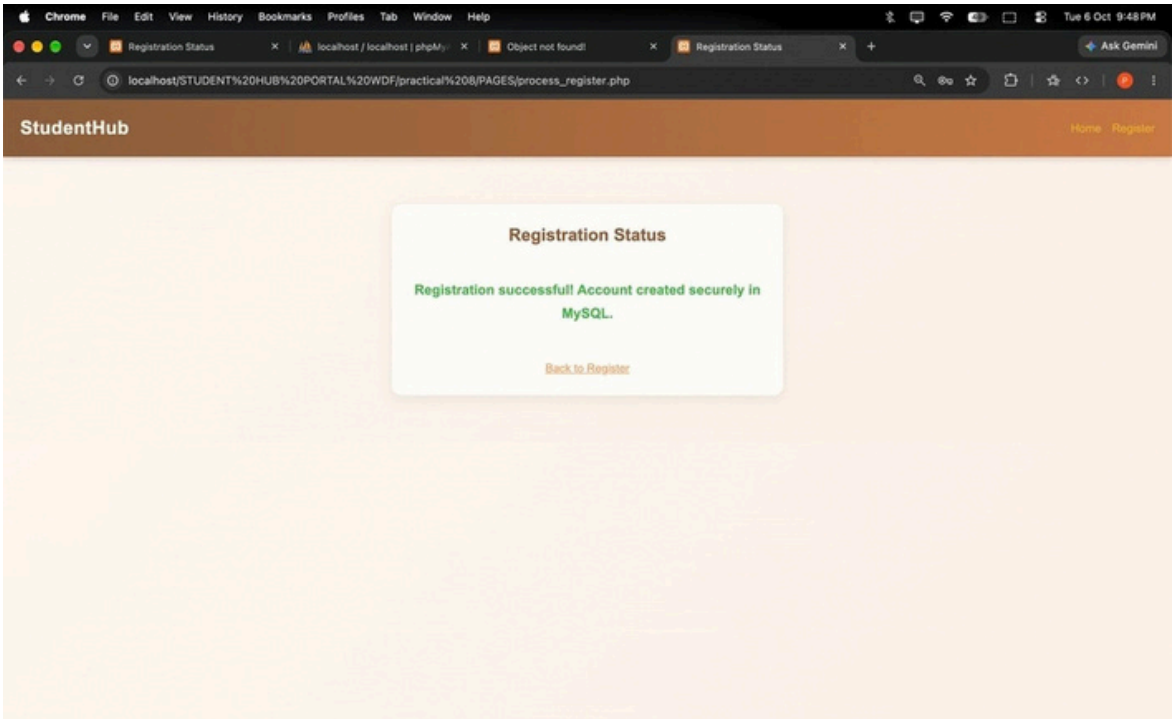
Is the database connection handled securely?

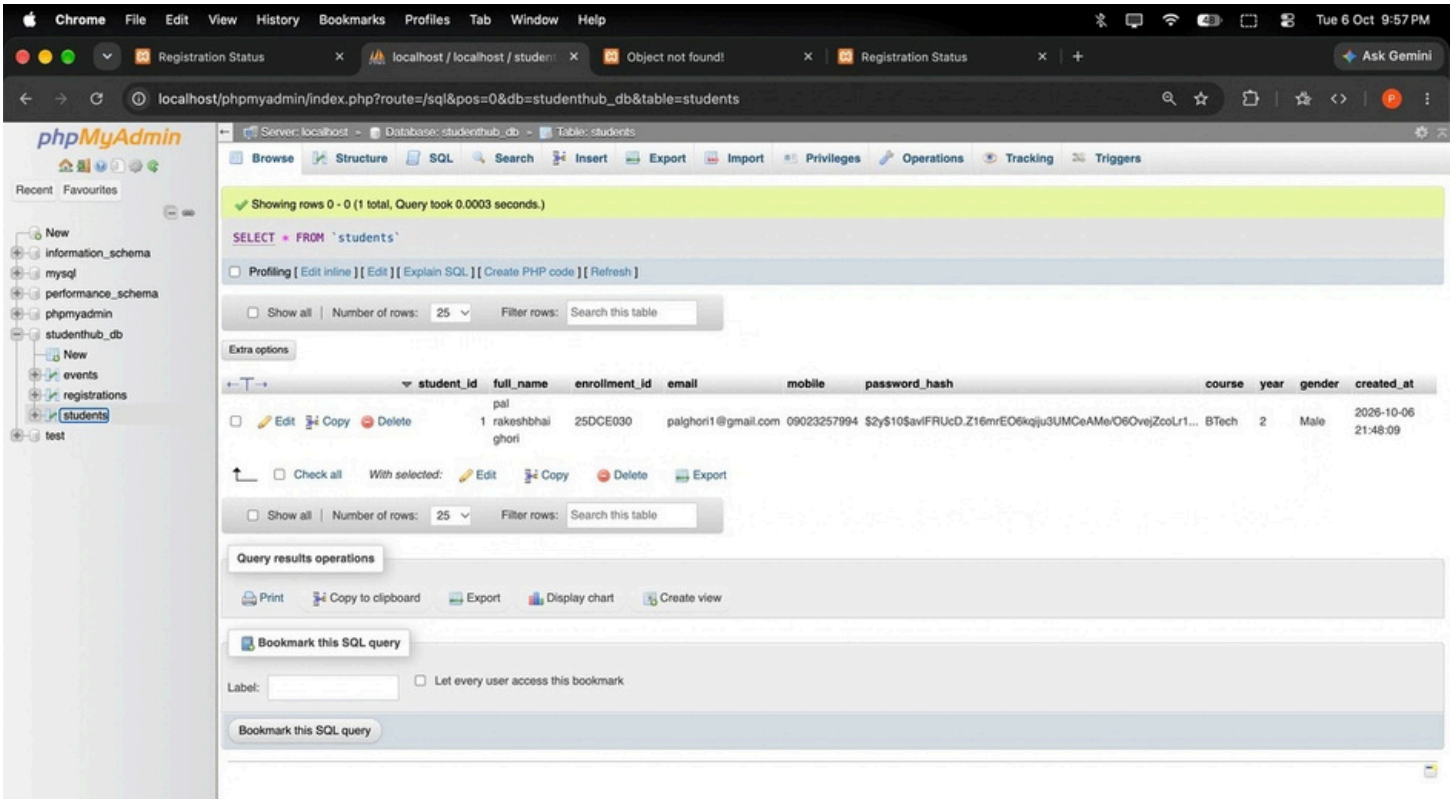
Yes. Practical 8 uses PDO, prepared statements, and prepare()/execute() instead of directly inserting user input into SQL queries. This helps prevent SQL Injection. Passwords are also stored using password_hash() rather than as plain text.

B. Implementation

- Created three MySQL tables: students, events, and registrations.
- Defined Primary Keys and Foreign Keys to establish relationships between the tables.
- Connected PHP with MySQL using PDO.
- Used prepared statements with prepare() and execute() to insert and retrieve data securely.
- Used password_hash() to store passwords securely instead of saving plain-text passwords.
- Stored student registration and event-related data in the relational database.

C. Evidence of Work (Screenshots)




Chrome File Edit View History Bookmarks Profiles Tab Window Help
Tue 6 Oct 9:57 PM
Registration Status
localhost / localhost / studen
Object not found!
Registration Status
Ask Gemini
localhost/phpmyadmin/index.php?route=/sql&pos=0&db=studenthub_db&table=students
phpMyAdmin
Recent Favourites
New
information_schema
mysql
performance_schema
phpmyadmin
studenthub_db
New
events
registrations
students
test
Server: localhost » Database: studenthub_db » Table: students
Browse Structure SQL Search Insert Export Import Privileges Operations Tracking Triggers
Showing rows 0 - 0 (1 total, Query took 0.0003 seconds.)
SELECT * FROM `students`
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]
Show all | Number of rows: 25 Filter rows: Search this table
Extra options
student_id full_name enrollment_id email mobile password_hash course year gender created_at
Edit Copy Delete 1 pal rakeshbhai ghori 25DCE030 palghori1@gmail.com 09023257994 $2y$10$avlFRUcD.Z16mrEO6kqjju3UMCeAMe/O6OvejZcoLr1... BTech 2 Male 2026-10-06 21:48:09
Check all With selected: Edit Copy Delete Export
Show all | Number of rows: 25 Filter rows: Search this table
Query results operations
Print Copy to clipboard Export Display chart Create view
Bookmark this SQL query
Label:
Let every user access this bookmark
Bookmark this SQL query

D. GitHub Repository Link:

https://github.com/palghori/STUDENT-HUB-PORTAL-WDF/tree/main/practical%208

E. Challenges Faced:

- Understanding the relationship between students, events, and registrations.
- Correctly defining Primary Keys and Foreign Keys.
- Handling database connection errors using PDO.
- Writing SQL queries with prepared statements and placeholders.
- Avoiding duplicate records and maintaining data consistency.
- Understanding the difference between password hashing and normal encryption.

F. Learning Outcomes:

- Learned how to design a basic relational database schema.
- Understood Primary Keys, Foreign Keys, and table relationships.
- Learned how PHP communicates with MySQL using PDO.
- Learned how prepared statements prevent SQL Injection.
- Learned secure password storage using password_hash().
- Gained practical knowledge of storing and managing dynamic website data in a real database instead of JSON/CSV files.

Faculty Remarks: